

Levels of Measurement: Nominal, Ordinal, Interval and Ratio

Levels of Measurement

Nominal	Ordinal	Interval	Ratio
"Eye color"	"Level of satisfaction"	"Temperature"	"Height"
Named	Named	Named	Named
	Natural order	Natural order	Natural order
		Equal interval between variables	Equal interval between variables
			Has a "true zero" value, thus ratio between values can be calculated

In statistics, we use data to answer interesting questions. But not all data is created equal. There are actually four different **data measurement scales** that are used to categorize different types of data:

1. Nominal
2. Ordinal
3. Interval
4. Ratio

Nominal

The simplest measurement scale we can use to label variables is a **nominal scale**. **Nominal scale:** A scale used to label variables that have no quantitative values.

Some examples of variables that can be measured on a nominal scale include:

- **Gender:** Male, female
- **Eye color:** Blue, green, brown
- **Hair color:** Blonde, black, brown, grey, other
- **Blood type:** O-, O+, A-, A+, B-, B+, AB-, AB+
- **Political Preference:** Republican, Democrat, Independent
- **Place you live:** City, suburbs, rural

Variables that can be measured on a nominal scale have the following properties:

- **They have no natural order.** For example, we can't arrange eye colors in order of worst to best or lowest to highest.
- **Categories are mutually exclusive.** For example, an individual can't have *both* blue and brown eyes. Similarly, an individual can't live *both* in the city and in a rural area.
- **The only number we can calculate for these variables are counts.** For example, we can count how many individuals have blonde hair, how many have black hair, how many have brown hair, etc.
- **The only measure of central tendency we can calculate for these variables is the mode.** The mode tells us which category had the most counts. For example, we could find which eye color occurred most frequently.

The most common way that nominal scale data is collected is through a survey. For example, a researcher might survey 100 people and ask each of them what type of place they live in.

Question: What type of area do you live in?

Possible Answers: City, Suburbs, Rural.

Using this data, the researcher can find out how many people live in each area, as well as which area is the most common to live in.

Ordinal

The next type of measurement scale that we can use to label variables is an **ordinal scale**.

Ordinal scale: A scale used to label variables that have a natural *order*, but no quantifiable difference between values.

Some examples of variables that can be measured on an ordinal scale include:

- **Satisfaction:** Very unsatisfied, unsatisfied, neutral, satisfied, very satisfied
- **Socioeconomic status:** Low income, medium income, high income
- **Workplace status:** Entry Analyst, Analyst I, Analyst II, Lead Analyst
- **Degree of pain:** Small amount of pain, medium amount of pain, high amount of pain

Variables that can be measured on an ordinal scale have the following properties:

- **They have a natural order.** For example, "very satisfied" is better than "satisfied," which is better than "neutral," etc.
- **The difference between values can't be evaluated.** For example, we can't exactly say that the difference between "very satisfied" and "satisfied" is the same as the difference between "satisfied" and "neutral."
- **The two measures of central tendency we can calculate for these variables are *the mode* and *the median*.** The mode tells us which category had the most counts and the median tells us the "middle" value.

Ordinal scale data is often collected by companies through surveys who are looking for feedback about their product or service. For example, a grocery store might survey 100 recent customers and ask them about their overall experience.

Question: How satisfied were you with your most recent visit to our store?

Possible Answers: Very unsatisfied, unsatisfied, neutral, satisfied, very satisfied.

Using this data, the grocery store can analyze the total number of responses for each category, identify which response was most common, and identify the median response.

Interval

The next type of measurement scale that we can use to label variables is an **interval scale**.

Interval scale: A scale used to label variables that have a natural order and a quantifiable difference between values, *but no "true zero" value*.

Some examples of variables that can be measured on an interval scale include:

- **Temperature:** Measured in Fahrenheit or Celsius
- **Credit Scores:** Measured from 300 to 850
- **SAT Scores:** Measured from 400 to 1,600

Variables that can be measured on an interval scale have the following properties:

- **These variables have a natural order.**
- **We can measure the mean, median, mode, and standard deviation of these variables.**
- **These variables have an exact difference between values.** Recall that ordinal variables have no exact difference between variables – we don't know if the difference between "very satisfied" and "satisfied" is the same as the difference between "satisfied" and "neutral." For variables on an interval scale, though, we know that the difference between a credit score of 850 and 800 is the exact same as the difference between 800 and 750.
- **These variables have no "true zero" value.** For example, it's impossible to have a credit score of zero. It's also impossible to have an SAT score of zero. And for temperatures, it's possible to have negative values (e.g. -10°F) which means there isn't a true zero value that values can't go below.

The nice thing about interval scale data is that it can be analyzed in more ways than nominal or ordinal data. For example, researchers could gather

data on the credit scores of residents in a certain county and calculate the following metrics:

- Median credit score (the “middle” credit score value)
- Mean credit score (the average credit score)
- Mode credit score (the credit score that occurs most often)
- Standard deviation of credit scores (a way to measure how spread out credit scores are)

Ratio

The last type of measurement scale that we can use to label variables is a **ratio scale**.

Ratio scale: A scale used to label variables that have a natural order, a quantifiable difference between values, and a “true zero” value.

Some examples of variables that can be measured on a ratio scale include:

- **Height:** Can be measured in centimeters, inches, feet, etc. and cannot have a value below zero.
- **Weight:** Can be measured in kilograms, pounds, etc. and cannot have a value below zero.
- **Length:** Can be measured in centimeters, inches, feet, etc. and cannot have a value below zero.

Variables that can be measured on a ratio scale have the following properties:

- **These variables have a natural order.**
- **We can calculate the mean, median, mode, standard deviation, and a variety of other descriptive statistics for these variables.**
- **These variables have an exact difference between values.**
- **These variables have a “true zero” value.** For example, length, weight, and height all have a minimum value (zero) that can’t be exceeded. It’s not possible for ratio variables to take on negative values. For this reason, the *ratio* between values can be calculated. For example, someone who weighs 200 lbs. can be said to weigh *two times* as much as someone who weighs 100 lbs. Likewise someone who is 6 feet tall is *1.5 times* taller than someone who is 4 feet tall.

Data that can be measured on a ratio scale can be analyzed in a variety of ways. For example, researchers could gather data about the height of individuals in a certain school and calculate the following metrics:

- Median height
- Mean height
- Mode height
- Standard deviation of heights
- Ratio of tallest height to smallest height

Summary

The following table provides a summary of the variables in each measurement scale:

Property	Nominal	Ordinal	Interval	Ratio
Has a natural "order"	YES	YES	YES	YES
Mode can be calculated	YES	YES	YES	YES
Median can be calculated		YES	YES	YES
Mean can be calculated			YES	YES
Exact difference between values			YES	YES
Has a "true zero" value				YES